Statistics and Econometrics with R

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July 2026

Contents

Welcome

This book grew out of two courses I teach at Azim Premji University — *Introduction to Statistics and Programming* and *Econometric Methods*.

In both courses I noticed the same thing. Students would learn a formula in one session and learn the R command for it in another, and the two never quite met. They could compute a t-statistic by hand and they could type `t.test()`, but the connection between the two stayed fuzzy.

This book is my attempt to fix that. **There is no separate "R chapter."** Every idea appears three times, in the same breath:

> **THE IDEA IN PLAIN WORDS**
>
> First, in plain words — what are we actually trying to find out, and why would anyone care?

> **WHERE THIS COMES FROM**
>
> Then, on paper — the algebra, worked slowly, with every step shown. You should be able to reproduce it in an exam with nothing but a pen.

> **DOING IT IN R**
>
> Then, in R — the same example, the same numbers, run as real code. Every result you see printed in this book was computed when the page was built. If the code were wrong, the number would be wrong, and I would notice.

How to use this book

Read it in order, at least the first time. Chapters build on each other; Part II assumes Part I.

Type the code out yourself. Copy-paste teaches you very little. Change a number and see what breaks — that is where the learning is.

Do the exercises before reading the solutions. They are worth nothing unless you have already tried.

What you need

You need two free programs, installed in this order:

1. **R** — the language itself, from cran.r-project.org
2. **RStudio Desktop** — a much friendlier window onto R, from posit.co/download/rstudio-desktop

A handful of chapters use extra packages. Install them once, by running this in the RStudio console:

```
install.packages(c("tidyverse", "BSDA", "wooldridge", "AER", "stargazer"))
```

You will not need anything else, and you will not need to pay for anything.

Notation used throughout

Symbol	Means
μ, σ	population mean, population standard deviation
$\bar{x}$, s	sample mean, sample standard deviation
n	sample size
$\hat{\beta}$	an *estimate* of the parameter β
α	significance level (the risk we accept of being wrong)
H_0, H_A	null hypothesis, alternative hypothesis

Greek letters are things about the **population** that we never observe. Roman letters with hats or bars are things we **compute from a sample** in the hope that they are close to the Greek ones. Almost all of statistics is the gap between those two columns.

A note on the data

Every dataset used in this book is either built into R, shipped alongside the book, or generated in front of you by code you can read. You will never see `read.csv(file.choose())` in these pages — that command needs a human to point at a file, which means nobody can reproduce your result, including you, six months from now.

> **REMEMBER THIS**
>
> If a result cannot be reproduced by running a script from top to bottom, it is not yet a result. It is a rumour.

One dataset deserves a word. The class survey was collected from students in this course, and originally carried their names. Those were replaced with anonymous ids before this book was ever published — which is itself the first lesson of working with data about people. You will meet the same idea again when we discuss what a sample is, and what you owe the people in it.

Licence and citation

If you spot an error, the source of every chapter is on GitHub and corrections are welcome as pull requests.

Corrections from students are genuinely welcome; several of the sharper points in this book exist because someone asked why.

To cite:

> Kulkarni, K. (2026). *Statistics and Econometrics with R.* Azim Premji University. https://kedarkulkarni1991.github.io/stats-econometrics-r/ Licensed under CC BY 4.0.

Chapter 1

Towards Inferential Statistics

Everything you have done with data so far has been **descriptive**. You take a set of numbers and summarise it: where the values sit, how far they spread, how often each one turns up. Those summaries tell you what the data look like, and they stop there. They make no claim about anything you did not measure.

Two kinds of summary do most of the work. Measures of **central tendency** — the mean, the median, the mode — say where the data are concentrated. The mean matters most of the three, because it balances every observation against the others and gives a natural point of reference. Measures of **dispersion** — the variance and the standard deviation — say how far individual observations sit from that reference. The mean tells you what is typical; the variance tells you how much any single observation is likely to depart from it. In the language of distributions, these are the first and second moments.

Numbers on their own hide things, so we also look. A histogram shows how often each range of values occurs, and makes symmetry, skewness and outliers visible at a glance. A bar chart does the same job for a variable that takes only a few distinct values. For a continuous variable, a smooth density curve captures the overall shape and makes two datasets easy to compare.

All of this is essential, and it is always the first thing to do with a new dataset. It is how you notice trends, catch errors, and get a feel for what you are holding. But notice what it cannot do. **Every descriptive statement is a statement about the data in front of you** — and the data in front of you is almost never the thing you actually care about.

You have a survey of two thousand households and you want to say something about the country. You have factory records for one year and you want to say something about Indian industry. You have a class of thirty-three students and you want to say something about young Indians. In each case the numbers you can compute describe your sample, while the question you want answered is about a population you did not observe.

That gap is the problem, and it comes with a specific difficulty. Draw a different sample from the same population and you will get different numbers — a different mean, a different standard deviation. Every statistic you compute is part fact about the population and part accident of who happened to end up in your sample. Looking at one sample on its own, you have no way of telling how much of your answer is which.

Inferential statistics is the set of tools for making that leap and being honest about it. It works by asking a question that sounds strange the first time you hear it: not "what does my sample say?" but "if I drew samples like this over and over, how much would the answer move around?" Once you can answer that, you can say how far to trust the one sample you actually have.

This unit builds that idea from the ground up. We begin with what a distribution is, and why it needs randomness to exist at all. We then meet the **sampling distribution** — the pattern formed not by raw observations but by a statistic computed from repeated samples — which is the pivot the whole subject turns on. The two units that follow take up the theorems that make it work: the **Law of Large Numbers**, which says the sample mean settles onto the population mean as samples grow, and the **Central Limit Theorem**, which says the shape of a sampling distribution is close to normal no matter what the population looks like.

1.1 Descriptive statistics, briefly

Before leaving description behind, it is worth seeing the tools on real data.

The Annual Survey of Industries records value added by registered factories in each Indian state:

```r
asi <- read.csv("data/asi-state-gva.csv")
asi_2010 <- subset(asi, yr == 2010 & !is.na(gva))

# ASI reports value added in rupees lakh; 100 lakh = 1 crore
asi_2010$gva_crore <- asi_2010$gva / 100

c(states = nrow(asi_2010),
  mean   = mean(asi_2010$gva_crore),
  median = median(asi_2010$gva_crore),
  sd     = sd(asi_2010$gva_crore))
```

```
#> states   mean median     sd
#>     32  25785  12752  36814
```

Nothing here is mysterious. We have taken 32 numbers and replaced them with four. That is **descriptive statistics**: the art of summarising the data you have so that a human can hold it in mind.

1.2 Where description runs out

To see the limit sharply, compare two questions that look identical.

First. What is the average height in this class?

```
students <- read.csv("data/stats-class.csv")

mean(students$height, na.rm = TRUE)
```

```
#> [1] 169.5
```

About 169.5 cm.

Now: how confident are you in that number?

> THE IDEA IN PLAIN WORDS
>
> Completely confident — and the question is almost meaningless. You measured everyone. The average height of this class is not an estimate of anything; it is a fact you computed. There is no margin of error. Asking "is it statistically significant?" would be incoherent: significantly different from what?

Second. What is the average industrial value added of an Indian state?

```
mean(asi_2010$gva_crore)
```

```
#> [1] 25785
```

Identical arithmetic. Completely different situation — for two distinct reasons, and it is worth separating them.

> WATCH OUT
>
> **The ASI is a survey, not a census.** It does not visit every factory; it samples them. Each state figure is already an estimate carrying sampling error, before we compute anything.
> **And what do you want the number *for*?** If you genuinely care only about

> those 32 states in that one year, the mean is a fact, like the class average. But nobody wants that. They want to say something about *Indian industry* — next year too, unregistered firms too, under a different policy too. The moment you want that, you have left description behind.

Inferential statistics is the discipline of making that leap and being honest about how far you have jumped.

> **REMEMBER THIS**
>
> Descriptive statistics summarises the data you have.
> Inferential statistics uses the data you have to make claims about data you do not have — and attaches a statement of how uncertain those claims are.
> A number without an honest account of its uncertainty is not a finding. It is an anecdote with decimal places.

Before we can make the leap, we must be clear about what we are leaping between. That requires the idea of a distribution.

1.3 What a distribution actually is

A **distribution** is a pattern that emerges from repetition under randomness. It tells you which outcomes are common, which are rare, and where values tend to concentrate.

The key word is *repetition*. A single observation has no distribution. Roll a die once and you get a four; that tells you almost nothing. To see the pattern you must roll again and again, recording outcomes, until the shape appears.

```r
set.seed(1)
rolls <- sample(1:6, size = 10000, replace = TRUE)

round(prop.table(table(rolls)), 3)
```

```
#> rolls
#>     1     2     3     4     5     6
#> 0.169 0.162 0.160 0.170 0.173 0.166
```

Each face appears roughly one-sixth of the time. That pattern *is* the distribution of a fair die, and it became visible only through repetition.

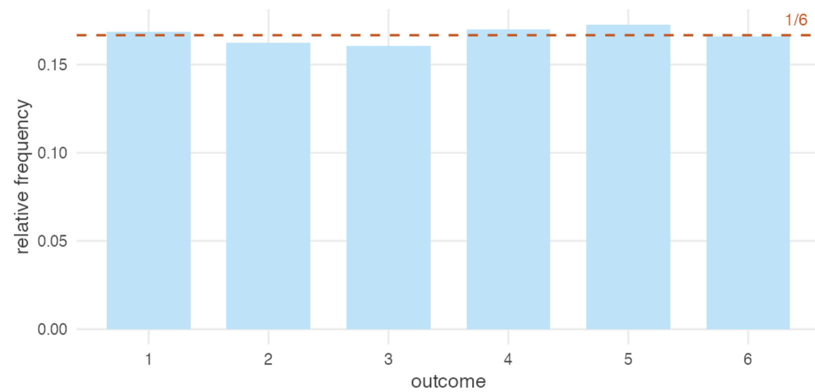

Figure 1.1: Ten thousand rolls of a fair die. Each face turns up close to one-sixth of the time.

1.3.1 Randomness is not optional

Now take the randomness away. Imagine a die with a 3 on every face.

```
rolls_fixed <- rep(3, 10000)

table(rolls_fixed)
```

```
#> rolls_fixed
#>     3
#> 10000
```

```
sd(rolls_fixed)
```

```
#> [1] 0
```

Every observation is identical. No spread, no uncertainty, standard deviation exactly zero. The distribution has collapsed onto a single point — what is called a **degenerate** distribution.

> REMEMBER THIS
>
> **A distribution is not a property of a single observation. It is a property of repeated observations generated by a random process.**
> Where there is no randomness there is nothing to describe, nothing to be uncertain about, and no statistics to do. Every technique in this book is a way of handling variation. Remove the variation and the entire subject

> falls silent.

This is why statistics is built on **random variables** — quantities whose value depends on the outcome of a random process. Height, income, rainfall, the number on a die: all vary, and it is precisely their variation that we model.

1.3.2 Expectation as a long-run average

If a random variable takes value x_i with probability $p(x_i)$, its **expected value** is

$$E[X] = \sum_i x_i \, p(x_i)$$

The name invites a misunderstanding. $E[X]$ is not the value you expect on any given try — for a fair die it is 3.5, a number the die cannot produce. It is the **long-run average** over many repetitions.

```
mean(rolls)                # average over 10,000 actual rolls
```

```
#> [1] 3.514
```

```
sum(1:6 * rep(1/6, 6))     # theoretical E[X]
```

```
#> [1] 3.5
```

> **THE IDEA IN PLAIN WORDS**
>
> That the two agree is neither a coincidence nor trivial. It is the first appearance of the Law of Large Numbers, which we will meet properly later: run a random process enough times and the average of what you observe closes in on the expectation you calculated on paper.
> Notice how unremarkable that felt. We are about to need it for something much less obvious.

1.4 Population, sample, and the assumption that licenses everything

Now the vocabulary of the leap.

The **population** is the entire collection you want to know about. Every registered factory in India. Every agricultural labourer. Every household eligible for a job guarantee scheme.

The **sample** is the part you actually observe.

A **parameter** is a number describing the population — the true mean μ, the true standard deviation σ. You never see these. They are written as Greek letters as a permanent reminder that they are out of reach.

A **statistic** is a number computed from your sample — $\bar{x}$, s. You can always see these. They are your only evidence.

> **REMEMBER THIS**
>
> The whole problem of inference, in one line:
>
> $$\underbrace{\bar{x}}_{\text{you have this}} \quad \longrightarrow \quad \underbrace{\mu}_{\text{you want this}}$$
>
> The arrow is not an equals sign. Making it trustworthy, and saying precisely how much to trust it, is what the rest of this book is for.

Why should a sample tell us anything at all about a population? Because of an assumption we make — and it is worth stating explicitly, because everything rests on it.

We assume the population has an underlying probability distribution, and that the members of our sample are **independent and identically distributed** (i.i.d.) draws from it.

- **Identically distributed**: every observation comes from the same distribution. No drift, no changing the rules partway through.
- **Independent**: knowing one observation tells you nothing about another.

> **REMEMBER THIS**
>
> It is this assumption — that the sample is drawn from the same distribution as the population — that licenses inference at all.
> If the sample came from a *different* distribution than the population, no amount of arithmetic could bridge them. This is why sampling design matters so much, and why a badly drawn sample cannot be repaired by clever statistics afterwards. The formulas will still run. They will simply be answering a question about a population you did not mean.

1.5 The pivot: distributions of statistics

Here is the conceptual step that makes inference possible, and it is the most important idea in this unit.

Everything so far concerned the distribution of **raw observations**: repeat an experiment, record individual outcomes, watch a pattern form. Now we shift attention from raw observations to **summaries of observations**.

Instead of recording one die roll at a time, roll the die thirty times, calculate the average, and record *only that average*. Then do the whole thing again: a fresh thirty rolls, a fresh average. And again, hundreds of times.

The pattern formed by those averages is a **sampling distribution**.

> **REMEMBER THIS**
>
> The distribution of a **random variable** describes how raw observations vary.
> A **sampling distribution** describes how a *statistic* — a sample mean, a proportion, a standard deviation — varies across repeated samples of a fixed size.
> It answers the question inference actually turns on: *if I took another sample, how different would my answer be?*

1.5.1 A worked example: a deck of cards

Take a standard deck as the population. Score the cards A = 1, J = 11, Q = 12, K = 13, and 2 through 10 at face value. Because we built this population ourselves, we know its parameters exactly — a luxury we will never have again.

```
deck <- rep(1:13, times = 4)

c(N = length(deck), mu = mean(deck), sigma = sd(deck))
```

```
#>      N     mu  sigma
#> 52.000  7.000  3.778
```

The population mean is exactly $\mu = 7$.

Now draw a sample of 10 cards and take its mean:

```
set.seed(7)
mean(sample(deck, size = 10, replace = TRUE))
```

```
#> [1] 5.7
```

WATCH OUT

Note `replace = TRUE`: each card is returned to the deck before the next draw.
That is not a technicality. It is what makes the draws **independent** — the i.i.d. assumption from the previous section. Draw *without* replacement and each card removed changes what remains, so the draws are dependent and the sample mean varies slightly less than the theory predicts. (The adjustment is called the finite population correction, and you can see it at work in the exercises.)
Sampling with replacement is also the better model of what we usually mean. When you survey 2,000 Indians you are not meaningfully depleting a population of a billion.

Not 7. Draw again and you would get something else again. **The sample mean is itself a random variable** — it has its own distribution, because it changes every time you draw.

So let us look at that distribution. Repeat the exercise a thousand times:

```
set.seed(7)
sample_means <- replicate(1000, mean(sample(deck, size = 10, replace = TRUE)))

c(mean_of_sample_means = mean(sample_means),
  sd_of_sample_means   = sd(sample_means))
```

```
#> mean_of_sample_means   sd_of_sample_means
#>                6.932                1.201
```

Two things happened, and both matter.

The sample means are centred on the population mean. No single sample got it right, but their average is 6.932 — essentially 7. This is the result $E(\bar{X}) = \mu$: the sample mean is an **unbiased** estimator. It is not right, but it is not *systematically* wrong, which is a different and far more useful property.

The sample means vary much less than the cards do. The population standard deviation is about 3.78; the sample means have a standard deviation of about 1.2. Averaging has cancelled out much of the noise.

1.5.2 What happens as the sample grows

That second observation is worth pushing on. What if we take bigger samples?

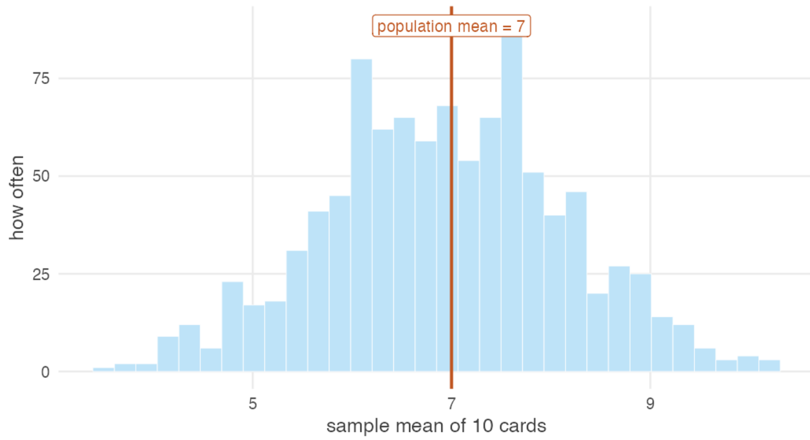

Figure 1.2: One thousand sample means, each from 10 cards. Individual samples miss the population mean constantly. Their average lands almost exactly on it.

```r
set.seed(7)

sizes <- c(5, 10, 20, 40, 80)
results <- t(sapply(sizes, function(n) {
  m <- replicate(2000, mean(sample(deck, size = n, replace = TRUE)))
  c(n = n, mean = mean(m), sd = sd(m), theoretical_sd = sd(deck) / sqrt(n))
}))

round(results, 3)
```

```
#>        n  mean    sd theoretical_sd
#> [1,]  5 6.932 1.688          1.690
#> [2,] 10 7.022 1.165          1.195
#> [3,] 20 7.009 0.845          0.845
#> [4,] 40 7.019 0.597          0.597
#> [5,] 80 6.996 0.415          0.422
```

Every row is centred on 7 — the sample mean is unbiased at any sample size. But the spread shrinks steadily, and in a completely predictable way. Compare the last two columns: the observed standard deviation of the sample means matches $\sigma/\sqrt{n}$ almost exactly.

> **REMEMBER THIS**
>
> Two results, which we will prove in a later unit:

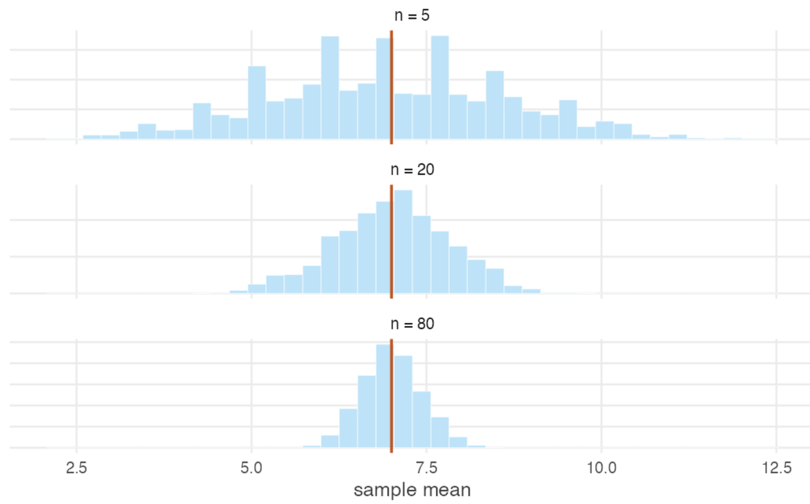

Figure 1.3: Sampling distributions of the mean for growing sample sizes. The centre never moves. The spread collapses.

$$E(\bar{X}) = \mu \qquad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

The sample mean is centred on the population mean, and its spread falls as the sample grows. The quantity $\sigma/\sqrt{n}$ is the **standard error**, and it is the single most important number in applied statistics: it is how wrong you should expect to be.

WATCH OUT

Note the square root. To halve your standard error you must **quadruple** your sample. Precision is expensive, and it grows more expensive the more of it you already have.
This is why surveys of two thousand people are common and surveys of two lakh people are not: the second costs a hundred times as much for about seven times the precision.

THE IDEA IN PLAIN WORDS

Look again at what just happened, because it is genuinely remarkable.
We began knowing nothing about how wrong a single sample would be.
By studying how the sample mean behaves *across many hypothetical sam-*

> *ples*, we worked out exactly how much it varies — and obtained a formula, $\sigma/\sqrt{n}$, that can be computed from a single real sample.
> In practice you only ever draw one sample. The sampling distribution tells you what would have happened had you drawn many, and that is enough to attach an honest margin of error to the one you have.

1.5.3 The same logic on Indian data

Cards are a clean laboratory. Real data is messier, and the ASI figures let us watch the same logic operate on something considerably less well behaved.

Treat the 32 state values as a population and sample from it:

```
set.seed(11)

pop <- asi_2010$gva_crore
asi_means <- replicate(2000, mean(sample(pop, size = 10, replace = TRUE)))

c(population_mean      = mean(pop),
  mean_of_sample_means = mean(asi_means),
  population_sd        = sd(pop),
  se_of_sample_means   = sd(asi_means))
```

```
#>      population_mean mean_of_sample_means        population_sd
#>                25785                25715                36814
#>   se_of_sample_means
#>                11543
```

Once again the sample means centre on the population mean, and once again they vary far less than the underlying data. The machinery does not care that industrial value added looks nothing like a deck of cards.

Which brings us to the feature of this data we have so far ignored.

1.6 The average is not the distribution

A habit worth forming now, before any technique gets in the way.

Almost all of inferential statistics is about averages. Nearly every method in this book estimates a mean, compares two means, or fits a line through conditional means. That focus is not arbitrary — means are well behaved, their sampling distributions are tractable, and the two great theorems are statements about them.

But the mean is only **one** feature of a distribution, and often not the one a policy question cares about.

```
c(mean   = mean(asi_2010$gva_crore),
  median = median(asi_2010$gva_crore))
```

```
#>   mean median
#>  25785  12752
```

```
mean(asi_2010$gva_crore) / median(asi_2010$gva_crore)
```

```
#> [1] 2.022
```

The mean is more than **twice** the median. When those two diverge that sharply, the average has stopped describing a typical state.

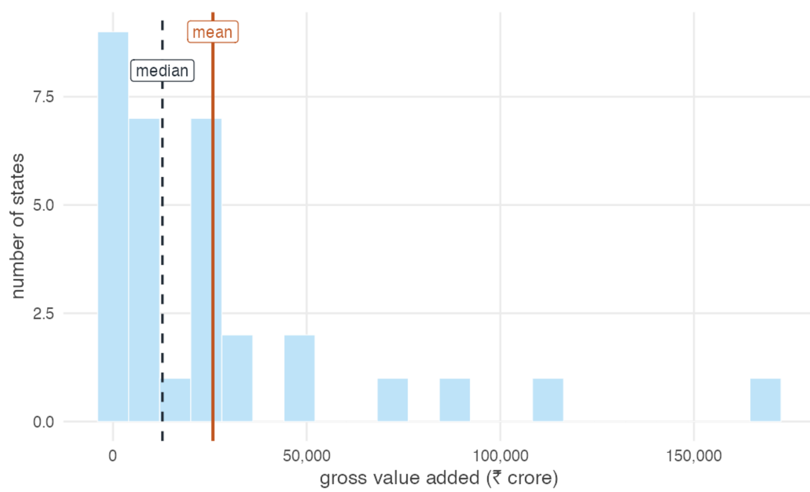

Figure 1.4: Industrial value added across Indian states, 2010. The mean sits far to the right of most states, dragged there by a handful of large ones.

How concentrated is it?

```
top5 <- sort(asi_2010$gva_crore, decreasing = TRUE)[1:5]

round(100 * sum(top5) / sum(asi_2010$gva_crore))              # % of all GVA
```

```
#> [1] 59
```

```r
round(max(asi_2010$gva_crore) / median(asi_2010$gva_crore), 1)   # top vs median
```

```
#> [1] 13.2
```

Five states hold about 59% of registered industrial value added. The largest produces roughly thirteen times the median state.

> **THE IDEA IN PLAIN WORDS**
>
> Now ask what "average industrial value added across states" is good for. Asking how much industry India has in total? Fine — the mean times the number of states is the total, by construction.
> Asking what industry looks like in a typical Indian state? Badly misleading. No state sits near the mean. The distribution is really two countries: a few industrial giants and a long tail of states doing something else.
> Asking whether industrial growth is spreading? The mean can rise while every state but one falls further behind. **An average can move in the opposite direction to almost everybody inside it.**

> **REMEMBER THIS**
>
> Ask two questions of every average you meet, including your own:
>
> 1. **What is the spread?** A mean with no measure of dispersion is half a sentence.
> 2. **Who is where?** Means are the wrong tool for questions about the bottom of a distribution — poverty, vulnerability, who is hurt by a shock. Those are questions about *tails*, and the mean is precisely the statistic that averages tails away.

This is not a philosophical aside; it changes what you estimate. Whole literatures in development economics exist because the mean was the wrong target. Poverty measures weight the poor rather than averaging over everyone. Vulnerability measures ask who is at risk of falling below a line. Distributional analysis asks whether a policy that raised average income raised the income of anyone who was badly off to begin with.

We will return to this repeatedly. When we fit a regression we will be modelling a conditional *mean*, and it will be worth asking each time what that choice conceals. When we reach binary explanatory variables and estimate an average treatment effect, the word "average" in that phrase will deserve every bit of suspicion you can bring to it.

WATCH OUT

None of this makes averages bad. They are the right tool for a great many questions, and this book spends most of its length on them for good reason.

It means an average is an **answer to a specific question**, not a neutral summary of the data. Decide which question you are asking before deciding the mean answers it.

1.7 What can go wrong before any mathematics

Inference assumes your sample carries information about the population you care about. Three ways that fails, none of which any formula will warn you about.

The sample is not from the population you mean. The ASI covers *registered* factories. Most Indian manufacturing employment is unregistered. Conclusions about "Indian manufacturing" drawn from ASI data are conclusions about a formal sliver of it. The arithmetic will not object.

The sample is not random. Our class survey has 32 heights. They are not a random sample of APU students, let alone of young Indians — they are the people who took this course and turned up that day. You may compute a confidence interval from them and it will mean nothing about the university.

The question is causal but the data is not. Suppose states with more industry have higher literacy. Does industry raise literacy? Does literacy attract industry? Does something else drive both? The correlation is identical in all three cases, and no statistical machinery separates them without an argument about *how the data came to exist*.

THE IDEA IN PLAIN WORDS

That last failure is why this book is organised as it is. From the unit on two-sample tests onwards, we stop asking only "what is this number?" and start asking "what would have to be true for this number to mean what I want it to mean?"

That question — not any particular estimator — is what separates a good applied economist from someone who can run regressions.

1.8 Summary

REMEMBER THIS

1. **Description** summarises data you have. **Inference** reaches beyond it and must report how far it has reached.
2. A **distribution** is a pattern emerging from repetition under randomness. No randomness, no distribution, no statistics.
3. Greek letters (μ, σ) are population parameters you never observe. Roman letters ($\bar{x}$, s) are sample statistics you compute.
4. Inference is licensed by the **i.i.d. assumption** — that your sample is drawn from the same distribution as the population. A badly drawn sample cannot be repaired later.
5. A **sampling distribution** describes how a statistic varies across repeated samples. It is the pivot on which all inference turns.
6. $E(\bar{X}) = \mu$ and $\mathrm{Var}(\bar{X}) = \sigma^2/n$. The sample mean is unbiased, and its **standard error** $\sigma/\sqrt{n}$ falls with the square root of the sample size.
7. **The mean is one feature of a distribution, not a summary of it.** Indian industrial value added has a mean twice its median, with five states holding 59% of the total. Always ask about spread, and about the tails.

We have watched sample means centre on the population mean and tighten as samples grow. We have not yet *proved* either claim, and we have said nothing about the **shape** of a sampling distribution.

Those are the next three units. First the distributions themselves — binomial, Poisson, normal, discrete against continuous — which describe a population completely and yet, on their own, still tell you nothing about how a *sample* from it will behave. Seeing exactly where they stop short is what makes the two theorems feel necessary rather than arbitrary.

Chapter 2

Hypothesis Testing: One Sample

Someone makes a claim about a population. You collect a sample. The sample disagrees with the claim — a little. Is that disagreement real, or is it just the luck of the draw?

That single question is what this chapter is about, and it is the engine underneath almost everything in the rest of the book.

2.1 The problem, before any formulas

The APU administration claims the average height of all students is 165 cm.

You are sceptical. You could settle it by measuring all 3,500 students, but that costs weeks of work. So instead you measure 36 students chosen at random, and you find their average height is **168 cm**.

The claim said 165. Your sample says 168. Is the administration wrong?

> **WATCH OUT**
>
> The tempting answer is "yes, obviously — 168 is not 165." That answer is wrong, and understanding *why* it is wrong is the whole chapter.
> You did not measure everyone. You measured 36 people. Even if the true average really were exactly 165, your particular 36 students would almost never average exactly 165. Some samples come out high, some low. **A gap between the sample and the claim is guaranteed to exist even when the claim is perfectly true.**

Let us actually watch that happen. Suppose we *know* the truth: the population mean is exactly 165 cm, with a standard deviation of 8 cm. Now let us draw a sample of 36 students, ten separate times, and look at the sample means.

```
set.seed(42)

ten_samples <- replicate(10, mean(rnorm(n = 36, mean = 165, sd = 8)))
round(ten_samples, 2)
```

```
#>  [1] 165.5 165.1 165.6 163.0 164.8 164.4 165.3 165.4 164.1 164.5
```

Every one of those numbers came from a population whose mean is *exactly* 165. Not one of them is 165. Several are further from 165 than a full centimetre.

So the real question is not "is 168 different from 165?" It obviously is. The real question is:

> THE IDEA IN PLAIN WORDS
>
> **If the administration's claim were true, how often would a sample of 36 students come out as far from 165 as ours did?**
> If the answer is "quite often" — then our sample is unremarkable, and we have no grounds to call anyone a liar.
> If the answer is "almost never" — then either something very unusual happened to us, or the claim is false. We go with the second explanation.

We can answer that question by brute force before we do any algebra at all. Let's draw not ten samples but fifty thousand, still assuming the claim is true, and count how many stray at least 3 cm away from 165 (as ours did).

```
set.seed(42)

many_means <- replicate(50000, mean(rnorm(n = 36, mean = 165, sd = 8)))

# How far did our real sample fall from the claim?
observed_gap <- abs(168 - 165)

mean(abs(many_means - 165) >= observed_gap)
```

```
#> [1] 0.02424
```

About 2.5% of the time. So if the administration is telling the truth, a sample like ours turns up roughly once in forty tries. That is rare enough to be suspicious — and by the end of this chapter you will be able to get that same number exactly, in one line, with no simulation at all.

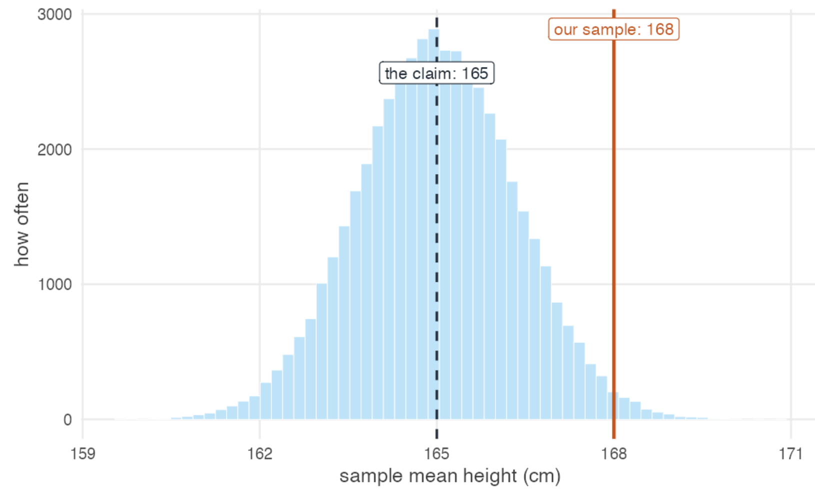

Figure 2.1: 50,000 sample means drawn from a population that truly averages 165 cm. Our observed sample mean of 168 sits far out in the tail.

2.2 Naming the two hypotheses

We start from a sample $x_1, x_2, \ldots, x_n$ drawn from a population with unknown mean μ and standard deviation σ. The claim under scrutiny is the **null hypothesis**:

$$H_0 : \mu = 165$$

The position we would move to if the claim collapses is the **alternative hypothesis**:

$$H_A : \mu \neq 165$$

REMEMBER THIS

The null hypothesis is always the *boring* option — nothing is going on, the claim stands, the difference is zero. We never *prove* it. We only ever fail to knock it down.
This is why we say "**fail to reject** H_0" and never "accept H_0." A court that acquits has not proven innocence; it has failed to prove guilt.

Table 2.1: Where the claim goes.

The question asks	Put the claim in	Because
”Can you **reject** the claim?”	H_0	You are testing whether the data car
”Is there evidence to **support** the claim?”	H_A	The claim is what you need evidenc

2.2.1 Which hypothesis does the claim go in?

This is where most marks are lost, so it is worth getting exactly right.

There is one structural rule, and it is not negotiable:

> **REMEMBER THIS**
>
> **The equals sign always lives in H_0.**
> H_0 gets $=$, $\leq$, or $\geq$. H_A gets $\neq$, $<$, or $>$ — never equality. This is not a convention; the test is *computed* assuming a specific value of μ, and only H_0 supplies one.

Given that, where the claim goes depends on what the question asks you to do with it:

> **WATCH OUT**
>
> A claim phrased with a **strict** inequality — “the mean is *more than* 499” — cannot go into H_0 as stated, because H_0 must contain the equality. If you are asked for evidence supporting that claim, it becomes
>
> $$H_0 : \mu \leq 499 \qquad H_A : \mu > 499$$
>
> Writing $H_0 : \mu \geq 499$ here is the common error. It puts the claim in the null *and* points the test in the wrong direction, so a sample mean **above** 499 — evidence *for* the claim — ends up producing a p-value near 1 and a “fail to reject.” The arithmetic is fine; the conclusion is backwards.

Work it through. A report claims the mean CAT score of applicants is *more than* 499. A sample of 100 has mean 502, with $\sigma = 10.6$. Is there enough evidence at $\alpha = 0.01$ to support the report?

The question says **support**, so the claim is the alternative:

$$H_0 : \mu \leq 499 \qquad H_A : \mu > 499$$

This is right-tailed. The critical value is $Z_{0.01} = 2.33$, not -2.33.

```r
xbar_c <- 502; mu_c <- 499; sigma_c <- 10.6; n_c <- 100

z_c <- (xbar_c - mu_c) / (sigma_c / sqrt(n_c))
c(z_calc = z_c,
  z_crit = qnorm(0.99),
  p_value = pnorm(z_c, lower.tail = FALSE))   # upper tail: H_A is ">"
```

```
#>   z_calc   z_crit  p_value
#> 2.830189 2.326348 0.002326
```

$Z_{calc} = 2.83 > 2.33$, and $p = 0.0023 < 0.01$. We **reject** H_0, which means the data **support** the report's claim.

THE IDEA IN PLAIN WORDS

Sanity-check the direction before you trust any p-value. The sample mean is 502 and the claim is "more than 499." The sample is *pointing the same way as the claim*, so the evidence must be favourable. If your arithmetic hands back $p = 0.998$ and "no support," you have pointed the tail the wrong way — go back to H_A.

2.3 How far is too far?

Write μ_0 for the value the null claims (here, 165). We will reject H_0 when the sample mean lands too far from μ_0 — that is, when

$$|\bar{x} - \mu_0| \geq c$$

for some cutoff c. Everything now hinges on choosing c.

WHERE THIS COMES FROM

We choose c so that, **if the null were true**, a gap that large would occur with probability only α — a small risk we decide in advance, typically 0.05:

$$P\big[|\bar{x} - \mu_0| \geq c\big] = \alpha$$

From the Central Limit Theorem (Chapter 2) we know $\bar{x}$ is normally distributed with mean μ and standard deviation $\sigma/\sqrt{n}$. Dividing through by that standard error turns the left-hand side into a standard normal:

$$P\left[\frac{|\bar{x}-\mu_0|}{\sigma/\sqrt{n}} \geq \frac{c}{\sigma/\sqrt{n}}\right] = \alpha$$

$$P\left[|Z| \geq \frac{c}{\sigma/\sqrt{n}}\right] = \alpha$$

The event $|Z| \geq k$ splits into two symmetric tails, each carrying half the probability:

$$2 \cdot P\left[Z \geq \frac{c}{\sigma/\sqrt{n}}\right] = \alpha \qquad \Longrightarrow \qquad P\left[Z \geq \frac{c}{\sigma/\sqrt{n}}\right] = \frac{\alpha}{2}$$

By definition $P[Z \geq Z_{\alpha/2}] = \alpha/2$, so the bracketed quantity *is* $Z_{\alpha/2}$, and therefore

$$\boxed{c = Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}}$$

Substituting that c back into the rejection rule and dividing both sides by the standard error gives the form you will actually use:

$$\frac{|\bar{x}-\mu_0|}{\sigma/\sqrt{n}} \geq Z_{\alpha/2}$$

The quantity on the left is the **test statistic**, written Z_{calc}. The value on the right is the **critical value**. The rule is simply: *if the statistic exceeds the critical value, reject the null.*

DOING IT IN R

Those critical values are the ones you have been reading off a printed table. There is no need — `qnorm()` gives you any of them to as many decimals as you like.

```r
alpha <- c(0.10, 0.05, 0.01)

# Two-tailed: we want the value leaving alpha/2 in the upper tail
qnorm(1 - alpha / 2)
```

```
#> [1] 1.645 1.960 2.576
```

Those three numbers are the 1.645, 1.96 and 2.58 you have memorised. Now you know where they come from, and you will never need the table again.

2.4 The worked example, both ways

Back to the students. We have $\bar{x} = 168$, $\mu_0 = 165$, $\sigma = 8$, $n = 36$, and we will work at $\alpha = 0.05$.

On paper

Step 1 — Hypotheses.

$$H_0 : \mu = 165 \qquad H_A : \mu \neq 165$$

Step 2 — Critical value. Two-tailed at $\alpha = 0.05$, so $Z_{0.025} = 1.96$.

Step 3 — Test statistic.

$$Z_{calc} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{168 - 165}{8/\sqrt{36}} = \frac{3}{8/6} = \frac{3}{1.333} = 2.25$$

Step 4 — Decision. $|2.25| > 1.96$, so we **reject** H_0 at the 5% level.

Step 5 — The p-value.

$$P(|Z| > 2.25) = 2 \times P(Z > 2.25) = 2 \times 0.01222 = 0.0244$$

If the administration's claim were true, only 2.4% of samples would sit as far from 165 as ours. That is unlikely enough to reject the claim.

The same thing in R

Now watch every one of those numbers reappear.

```
xbar  <- 168
mu0   <- 165
sigma <- 8
n     <- 36

se     <- sigma / sqrt(n)          # standard error
z_calc <- (xbar - mu0) / se        # the test statistic
z_crit <- qnorm(1 - 0.05 / 2)      # the critical value
p_val  <- 2 * pnorm(-abs(z_calc))  # the p-value

c(se = se, z_calc = z_calc, z_crit = z_crit, p_value = p_val)
```

```
#>      se  z_calc  z_crit p_value
#> 1.33333 2.25000 1.95996 0.02445
```

The standard error is 1.3333, the test statistic is 2.25, and the p-value is 0.0244 — exactly the hand calculation, and near enough exactly the 0.0242 we got by brute-force simulation at the start of the chapter.

THE IDEA IN PLAIN WORDS

Notice what just happened. The simulation, the algebra, and the R code are three descriptions of one idea. The formula is not a rule handed down from above — it is a shortcut for the counting exercise we did by hand in Section 2.1.

And here is the picture the numbers are describing:

```
plot_rejection(stat = 2.25, crit = 1.96, tails = "two",
               stat_label = "Z = 2.25")
```

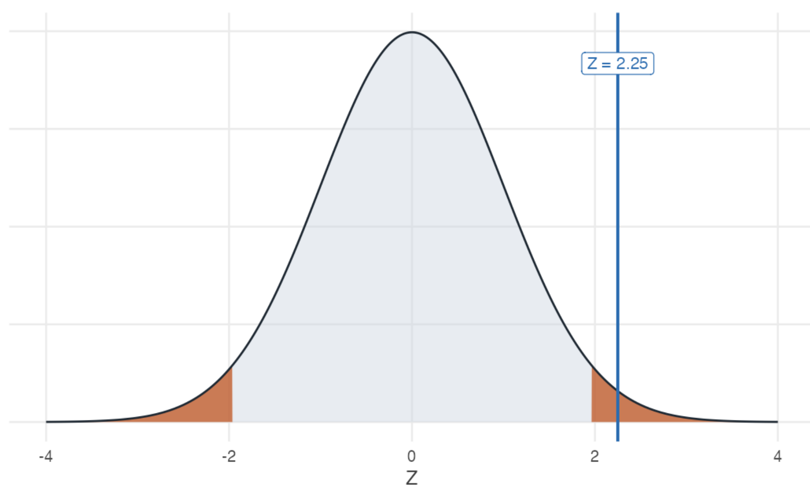

Figure 2.2: The rejection region at $\alpha = 0.05$ (shaded), with our test statistic of 2.25 falling inside it.

Letting a package do it

For a real dataset rather than summary statistics, `z.test()` from the **BSDA** package does the whole thing at once. Since our problem quoted only summary statistics, we first build a sample consistent with them:

```r
library(BSDA)

heights <- make_sample(n = 36, mean = 168, sd = 8)

z.test(heights, mu = 165, sigma.x = 8, conf.level = 0.95)
```

```
#>
#>  One-sample z-Test
#>
#> data:  heights
#> z = 2.3, p-value = 0.02
#> alternative hypothesis: true mean is not equal to 165
#> 95 percent confidence interval:
#>  165.4 170.6
#> sample estimates:
#> mean of x
#>       168
```

Read that output against the hand calculation: `z = 2.25`, `p-value = 0.0244`, and a confidence interval that we are about to explain.

> **WATCH OUT**
>
> `z.test()` needs `sigma.x` — the **population** standard deviation. If you do not know it (and in real work you almost never do), you must not use a Z-test. Use `t.test()` instead; see Section 2.7.

2.5 What a p-value actually says

The p-value is the single most misquoted number in empirical work, so let us be exact.

> **REMEMBER THIS**
>
> The p-value is the probability of seeing a sample **at least as extreme as ours**, *given that the null hypothesis is true*.
> It is **not** the probability that the null is true. It is **not** the probability that we made a mistake. It is a statement about the data, computed under an assumption about the world — not a statement about the world.

Small p-value means the data would be surprising if the null held, which is evidence against the null. Large p-value means the data are unremarkable under the null, so we have no case.

Table 2.2: Reading a p-value.

p-value	Read it as	At alpha = 0.05
0.60	Completely ordinary under the null.	Fail to reject
0.18	A bit high, but this happens roughly 1 time in 6.	Fail to reject
0.049	Would happen about 1 time in 20. Reject at 5%.	Reject
0.0001	Would happen 1 time in 10,000. Strong evidence.	Reject

2.6 Confidence intervals are the same test in disguise

Rearranging the "do not reject" condition $|\bar{x} - \mu| \leq Z_{\alpha/2}\,\sigma/\sqrt{n}$ for μ gives

$$\bar{x} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \;\leq\; \mu \;\leq\; \bar{x} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

For our students:

```
xbar + c(-1, 1) * qnorm(0.975) * (8 / sqrt(36))
```

```
#> [1] 165.4 170.6
```

The 95% interval is roughly $[165.4,\ 170.6]$. The claimed value of 165 lies **outside** it — which is the same verdict we reached with the test statistic, and it is no coincidence.

> **REMEMBER THIS**
>
> A $(1-\alpha)$ confidence interval is exactly the set of values of μ_0 that a two-tailed test at level α would **fail** to reject.
> Test and interval are one object viewed from two sides. If the interval misses the claimed value, the test rejects it. Always.

> **WATCH OUT**
>
> "95% confident" does **not** mean there is a 95% probability that μ lies in this particular interval. μ is a fixed number; it is either in there or it is not. It means: *the procedure* that generated this interval captures the true μ in 95% of repeated samples. The randomness lives in the interval, not in μ.

We can see that directly. Build 100 intervals from 100 samples of a population we know averages 165, and count the misses:

```
set.seed(1)

sims <- t(replicate(100, {
  s <- rnorm(36, mean = 165, sd = 8)
  mean(s) + c(-1, 1) * qnorm(0.975) * 8 / sqrt(36)
}))

ci <- data.frame(id = 1:100, lo = sims[, 1], hi = sims[, 2])
ci$misses <- ci$lo > 165 | ci$hi < 165

sum(ci$misses)   # how many of the 100 failed to capture the truth?
```

```
#> [1] 5
```

Five of the hundred missed. Run it again with a different seed and you will get four, or seven, or three — but over the long run, 5%.

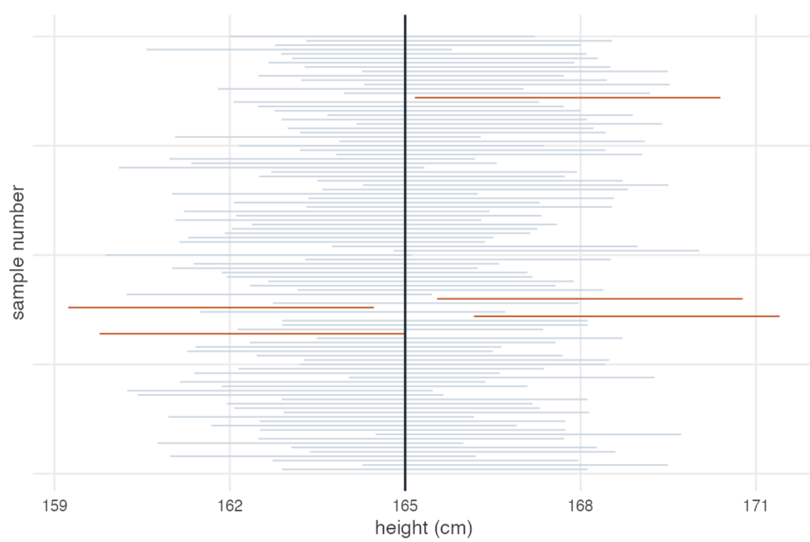

Figure 2.3: One hundred 95% confidence intervals from a population whose true mean is 165 (the vertical line). The intervals shown in orange are the ones that missed.

2.7 When σ is unknown: the t-test

Everything so far assumed we knew the population standard deviation σ. We almost never do. Replacing it with the sample standard deviation s adds a second source of uncertainty, and the statistic

$$t_{calc} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

no longer follows a normal distribution. It follows Student's t with $n-1$ **degrees of freedom** — a distribution with fatter tails, which demands more evidence before it will let you reject.

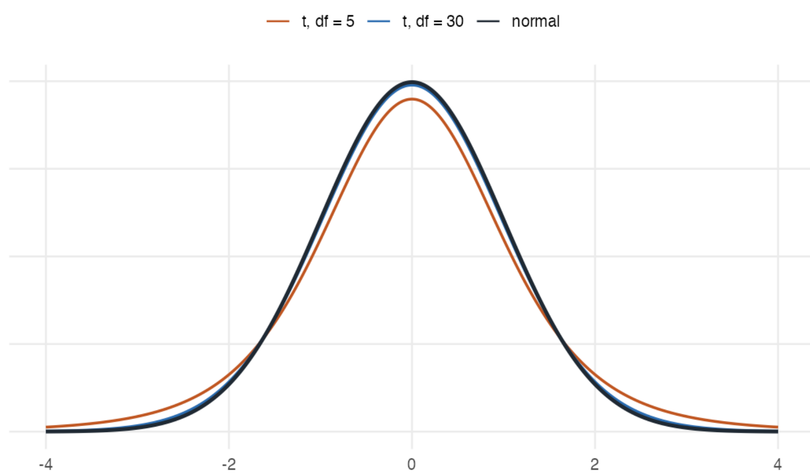

Figure 2.4: Student's t against the standard normal. With few degrees of freedom the tails are noticeably heavier, so critical values sit further out.

Settling the question with the actual class

We opened this chapter with a made-up sample of 36 students. We can now do the real thing. `stats-class.csv` holds the survey this class filled in — names, ages, heights, gender:

```
students <- read.csv("data/stats-class.csv")

str(students)
```

```
#> 'data.frame':    33 obs. of  4 variables:
#>  $ student_id: chr  "S01" "S02" "S03" "S04" ...
#>  $ age       : int  31 19 18 18 19 18 18 21 18 19 ...
#>  $ height    : int  178 161 181 164 170 175 176 186 161 170 ...
#>  $ gender    : chr  "M" "F" "M" "F" ...
```

Thirty-three rows, but look at the last one. One student left age and height blank, and another did not give a gender:

```
students[!complete.cases(students), ]
```

```
#>    student_id age height gender
#> 8         S08  21    186   <NA>
#> 33        S33  NA     NA      F
```

> **WATCH OUT**
>
> **A blank is not always an `NA`.** In a numeric column an empty field is read as `NA`, but in a text column it is read as the empty string `""` — and `""` is a perfectly ordinary value as far as `is.na()`, `complete.cases()` and `na.rm` are concerned. A gap in a text column can therefore hide in plain sight. If a dataset you did not prepare yourself has blanks, say so explicitly when you read it:
>
> ```
> read.csv("somefile.csv", na.strings = c("", "NA"))
> ```
>
> The files in this book are already written so that plain `read.csv()` does the right thing.

> **WATCH OUT**
>
> `mean()` returns `NA` if even one value is missing — it refuses to guess what you meant. Pass `na.rm = TRUE` to drop the gaps, and always know how many you dropped.
>
> ```
> mean(students$height)                  # NA — one height is missing
> ```
>
> ```
> #> [1] NA
> ```
>
> ```
> mean(students$height, na.rm = TRUE)    # 32 students
> ```
>
> ```
> #> [1] 169.5
> ```

Now the test. We do **not** know the population standard deviation of student heights, so this is a t-test, not a Z-test:

```
heights_real <- students$height[!is.na(students$height)]

c(n = length(heights_real),
```

```
  mean = mean(heights_real),
  sd = sd(heights_real))
```

```
#>      n   mean     sd
#>  32.00 169.50   8.71
```

```
t.test(heights_real, mu = 165)
```

```
#>
#>  One Sample t-test
#>
#> data:  heights_real
#> t = 2.9, df = 31, p-value = 0.006
#> alternative hypothesis: true mean is not equal to 165
#> 95 percent confidence interval:
#>  166.4 172.6
#> sample estimates:
#> mean of x
#>     169.5
```

$t = 2.92$ on 31 degrees of freedom, with a p-value of 0.0064. We **reject** the claim that the average is 165 cm. The 95% interval, $[166.4,\ 172.6]$, excludes 165 — the same verdict, as always.

> **THE IDEA IN PLAIN WORDS**
>
> Notice how much narrower this is than the textbook version of the problem. The opening example assumed we somehow knew $\sigma = 8$. Here we had to estimate it from 32 people, and the t distribution charges us for that uncertainty with a wider interval and a higher bar to clear.
> Also notice what the data is not: 32 students in one classroom are not a random sample of all APU students. The arithmetic is correct; whether it answers the question about *the university* is a separate matter, and a harder one.

Worked example

The dean estimates that full-time faculty teach 11.0 classroom hours per week. You sample eight faculty members. Can you reject the dean's claim at $\alpha = 0.10$? At $\alpha = 0.01$?

```r
hours <- c(11.8, 8.6, 10.5, 7.9, 6.4, 10.4, 10.0, 8.2)

c(n = length(hours), mean = mean(hours), sd = sd(hours))
```

```
#>     n  mean    sd
#> 8.000 9.225 1.749
```

On paper. With $\bar{x} = 9.225$, $s = 1.749$, $n = 8$ and $df = 7$:

$$t_{calc} = \frac{9.225 - 11}{1.749/\sqrt{8}} = \frac{-1.775}{0.618} = -2.87$$

The critical values are $t_{0.05,7} = 1.895$ and $t_{0.005,7} = 3.499$.

- At $\alpha = 0.10$: $2.87 > 1.895 \Rightarrow$ **reject** H_0.
- At $\alpha = 0.01$: $2.87 < 3.499 \Rightarrow$ **fail to reject** H_0.

In R. Both the pieces and the packaged version:

```r
t_calc <- (mean(hours) - 11) / (sd(hours) / sqrt(length(hours)))
t_crit <- qt(c(0.95, 0.995), df = 7)     # 10% and 1%, two-tailed

c(t_calc = t_calc, crit_10pct = t_crit[1], crit_1pct = t_crit[2])
```

```
#>     t_calc crit_10pct  crit_1pct
#>     -2.870      1.895      3.499
```

```r
t.test(hours, mu = 11, conf.level = 0.90)
```

```
#> 
#>  One Sample t-test
#> 
#> data:  hours
#> t = -2.9, df = 7, p-value = 0.02
#> alternative hypothesis: true mean is not equal to 11
#> 90 percent confidence interval:
#>   8.053 10.397
#> sample estimates:
#> mean of x 
#>     9.225
```

The p-value is 0.024 — below 0.10 but above 0.01, exactly matching the two different verdicts we reached by hand.

> **THE IDEA IN PLAIN WORDS**
>
> The same data can be "significant" at one level and not at another. Nothing has changed about the world between those two lines — only how much risk of a false alarm we were willing to tolerate. This is why reporting the p-value itself is far more informative than reporting a bare "significant / not significant."

2.8 One-tailed tests

Sometimes only one direction matters. A firm claims its new curing technique gets nicotine content *below* 1.5 mg; we only care if the true mean is lower.

$$H_0 : \mu \geq 1.5 \qquad H_A : \mu < 1.5$$

All the probability we are willing to risk now goes into a single tail, so the critical value moves *inward*: $-Z_{0.05} = -1.645$ rather than ± 1.96.

```r
qnorm(0.05)             # left-tailed critical value at 5%
```

```
#> [1] -1.645
```

```r
qnorm(0.95)             # right-tailed critical value at 5%
```

```
#> [1] 1.645
```

In R, add the `alternative` argument:

```r
nicotine <- make_sample(n = 20, mean = 1.42, sd = 0.7)

z.test(nicotine, mu = 1.5, sigma.x = 0.7, alternative = "less",
       conf.level = 0.95)
```

```
#> 
#>  One-sample z-Test
#> 
#> data:  nicotine
#> z = -0.51, p-value = 0.3
#> alternative hypothesis: true mean is less than 1.5
#> 95 percent confidence interval:
#>     NA 1.677
```

```
#> sample estimates:
#> mean of x
#>      1.42
```

$Z_{calc} = -0.51$, which is not below -1.645, so we fail to reject. The firm's claim is not supported — a sample mean of 1.42 would arise about 30% of the time even if the true mean were 1.5.

The one-sided interval looks backwards

This trips up nearly everyone. For the left-tailed test above, the confidence interval is

$$\left(-\infty,\ \bar{x} + Z_\alpha \frac{\sigma}{\sqrt{n}}\right)$$

An *upper* bound — for a *left*-tailed test. Why?

WHERE THIS COMES FROM

Start from the rejection rule and solve for μ_0 rather than for $\bar{x}$:

$$\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} < -Z_\alpha$$

$$\bar{x} - \mu_0 < -Z_\alpha \frac{\sigma}{\sqrt{n}}$$

$$\mu_0 > \bar{x} + Z_\alpha \frac{\sigma}{\sqrt{n}}$$

The inequality flipped when μ_0 moved across. So we reject exactly when μ_0 is **too large**, and the values we *fail* to reject are those with $\mu \leq \bar{x} + Z_\alpha\, \sigma/\sqrt{n}$.

THE IDEA IN PLAIN WORDS

The rejection region is defined in Z**-space**; the confidence interval lives in **parameter space**. Solving the inequality for μ reverses the direction. Put plainly: a left-tailed test rejects when the sample mean comes out too small. So the values of μ it rules out are the ones that are too *large* relative to what we saw. That leaves an upper bound.

Table 2.3: One-sample tests at a glance. Use Z and ‘z.test()‘ when σ is known; use t with $df = n - 1$ and ‘t.test()‘ when it is not.

Test	Hypotheses	Reject when	I
Two-tailed	$H_0: \mu = \mu_0$ vs $H_A: \mu \neq \mu_0$	$\|Z_{calc}\| > Z_{\alpha/2}$	(
Left-tailed	$H_0: \mu \geq \mu_0$ vs $H_A: \mu < \mu_0$	$Z_{calc} < -Z_{\alpha}$	‘
Right-tailed	$H_0: \mu \leq \mu_0$ vs $H_A: \mu > \mu_0$	$Z_{calc} > Z_{\alpha}$	‘

Table 2.4: Standard normal critical values, computed by ‘qnorm()‘ rather than copied from a table.

α	Two-tailed ($Z_{\alpha/2}$)	One-tailed (Z_α)
10%	1.645	1.282
5%	1.960	1.645
1%	2.576	2.326

2.9 Reference tables

2.10 Practice problems

YOUR TURN

Work each one on paper first, then check yourself in R. The solutions are one click away, which makes them worth nothing unless you try first. These are for learning, so the workings are shown in full. The question bank that follows is for testing yourself, and has no answers.

11.1 A consumer research organisation states that the mean caffeine content per 12-ounce bottle of a caffeinated soft drink is 37.7 mg. A random sample of 36 bottles has a mean of 35.28 mg. The population standard deviation is 10.8 mg. At $\alpha = 0.01$, can you reject the organisation’s claim?

SOLUTION

Solution. $H_0 : \mu = 37.7$ against $H_A : \mu \neq 37.7$. Two-tailed at $\alpha = 0.01$, so the critical value is $Z_{0.005} = 2.58$.

$$Z_{calc} = \frac{35.28 - 37.7}{10.8/\sqrt{36}} = \frac{-2.42}{1.8} = -1.34$$

```
caffeine <- make_sample(n = 36, mean = 35.28, sd = 10.8)
z.test(caffeine, mu = 37.7, sigma.x = 10.8, conf.level = 0.99)
```

```
#>
#>  One-sample z-Test
#>
#> data:  caffeine
#> z = -1.3, p-value = 0.2
#> alternative hypothesis: true mean is not equal to 37.7
#> 99 percent confidence interval:
#>  30.64 39.92
#> sample estimates:
#> mean of x
#>     35.28
```

$|-1.34| < 2.58$ and the p-value of 0.179 is far above 0.01, so we **fail to reject**. If the true mean really were 37.7 mg, a sample mean this far away would turn up about 18% of the time — thoroughly unremarkable.

11.2 Historical data show household water use is normally distributed with mean 360 gallons and standard deviation 40 gallons per day. A sample of 200 households averages 374 gallons. Are these data consistent with the historical distribution at the 5% level? What is the p-value?

SOLUTION

Solution.

$$Z_{calc} = \frac{374 - 360}{40/\sqrt{200}} = \frac{14}{2.828} = 4.95$$

```
water <- make_sample(n = 200, mean = 374, sd = 40)
z.test(water, mu = 360, sigma.x = 40, conf.level = 0.95)
```

```
#>
#>  One-sample z-Test
#>
#> data:  water
#> z = 4.9, p-value = 0.0000007
#> alternative hypothesis: true mean is not equal to 360
#> 95 percent confidence interval:
#>  368.5 379.5
#> sample estimates:
#> mean of x
#>       374
```

$4.95 \gg 1.96$, and the p-value is about 0 — under one in a million. **Reject** H_0. Water use has moved away from the historical mean.

11.3 A study claims the average monthly salary of full professors in India is ₹87,800. Students at one university sample ten salaries (in thousands):

```
91.0, 79.8, 102.0, 93.5, 82.0, 88.6, 90.0, 98.6, 101.0, 84.0
```

They believe their professors earn *more*. Test at the 10% and 5% levels.

SOLUTION

Solution. The word "more" makes this **right-tailed**: $H_0 : \mu \leq 87.8$ against $H_A : \mu > 87.8$. Since σ is unknown and $n = 10$, use a t-test with $df = 9$.

```r
salary <- c(91.0, 79.8, 102.0, 93.5, 82.0, 88.6, 90.0, 98.6, 101.0, 84.0)

c(mean = mean(salary), sd = sd(salary))
```

```
#>   mean     sd
#> 91.050  7.797
```

```r
t.test(salary, mu = 87.8, alternative = "greater")
```

```
#>
#>  One Sample t-test
#>
#> data:  salary
#> t = 1.3, df = 9, p-value = 0.1
#> alternative hypothesis: true mean is greater than 87.8
#> 95 percent confidence interval:
#>  86.53   Inf
#> sample estimates:
#> mean of x
#>     91.05
```

$t_{calc} = 1.318$, against critical values $t_{0.10,9} = 1.383$ and $t_{0.05,9} = 1.833$. The statistic falls short of both. **Fail to reject** at either level — there is not enough evidence that these professors are paid above the national figure.

11.4 A Department of Transportation claims mean wait time is *at most* 6 minutes. A sample of 34 locations has mean 10.3 minutes, standard deviation 8.0. Test at 5% and 1%.

SOLUTION

Solution. "At most" makes this right-tailed: $H_0 : \mu \leq 6$ against $H_A : \mu > 6$, t-test with $df = 33$.

```
waits <- make_sample(n = 34, mean = 10.3, sd = 8.0)
t.test(waits, mu = 6, alternative = "greater")
```

```
#>
#>  One Sample t-test
#>
#> data:  waits
#> t = 3.1, df = 33, p-value = 0.002
#> alternative hypothesis: true mean is greater than 6
#> 95 percent confidence interval:
#>  7.978   Inf
#> sample estimates:
#> mean of x
#>      10.3
```

$t_{calc} = 3.14$, well beyond $t_{0.05,\,33} = 1.692$ and $t_{0.01,\,33} = 2.445$. **Reject at both levels** — wait times are longer than claimed.

11.5 `marriage-ages.csv` records the ages of both partners in ten couples. Is there evidence that one partner tends to be older? Test at the 5% level.

Hint: the two columns are not independent samples — they are ten couples. What is the one number per couple that captures the question?

SOLUTION

Solution. The two columns are **paired**: each row is one couple. Treating them as two independent samples throws away that pairing and inflates the standard error. The right move is to reduce each couple to a single number — the age gap — and run a **one-sample** test on it against zero.

```
couples <- read.csv("data/marriage-ages.csv")

gap <- couples$partner1 - couples$partner2
gap
```

```
#>  [1] 1 5 4 1 1 4 0 8 1 0
```

```
c(mean = mean(gap), sd = sd(gap), n = length(gap))
```

```
#>   mean     sd      n
```

```
#>  2.500  2.635 10.000
```

```r
t.test(gap, mu = 0)
```

```
#>
#>  One Sample t-test
#>
#> data:  gap
#> t = 3, df = 9, p-value = 0.01
#> alternative hypothesis: true mean is not equal to 0
#> 95 percent confidence interval:
#>  0.6149 4.3851
#> sample estimates:
#> mean of x
#>       2.5
```

The mean gap is 2.5 years, $t = 3.0$ on 9 degrees of freedom, $p = 0.015$. Since $0.015 < 0.05$, we **reject** H_0: partner 1 is reliably older.
R will do the pairing for you if you ask:

```r
t.test(couples$partner1, couples$partner2, paired = TRUE)
```

```
#>
#>  Paired t-test
#>
#> data:  couples$partner1 and couples$partner2
#> t = 3, df = 9, p-value = 0.01
#> alternative hypothesis: true mean difference is not equal to 0
#> 95 percent confidence interval:
#>  0.6149 4.3851
#> sample estimates:
#> mean difference
#>             2.5
```

Identical output — `paired = TRUE` computes exactly the differences we formed by hand. Compare what happens if you forget:

```r
t.test(couples$partner1, couples$partner2, paired = FALSE)$p.value
```

```
#> [1] 0.607
```

The same ten couples, a p-value of 0.607 instead of 0.015, and the opposite conclusion. **Pairing is not a detail.**

11.6 *(No computation.)* A classmate reports "$p = 0.03$, so there is a 3% chance the null hypothesis is true." Explain what is wrong, and state what $p = 0.03$ does mean.

SOLUTION

Solution. The p-value is computed **assuming the null is true**. It cannot then also be the probability that the null is true — that would be circular. Formally, your classmate has swapped $P(\text{data} \mid H_0)$ for $P(H_0 \mid \text{data})$, which are different quantities entirely.
The correct reading: *if* H_0 were true, samples at least as extreme as this one would arise 3% of the time. Since that is unusual, we treat it as evidence against H_0 — but the statement is about how surprising the data are, not about how probable the hypothesis is.

2.11 Question bank

WATCH OUT

No solutions are given for this section, by design. These questions are drawn from past quizzes and exams. If you can work them without checking an answer key, you are ready; if you cannot, the section to reread is named beside each one.
They run from easiest to hardest. Do them in order.

Multiple choice

QB 11.1 The level of significance, α, is the risk of *(select all that apply)*

a. Rejecting the null hypothesis when it is true
b. Rejecting the null hypothesis when the alternative hypothesis is true
c. Rejecting the alternative hypothesis when the null hypothesis is true
d. Rejecting the alternative hypothesis when it is true

See Section 2.3.

QB 11.2 An economist sets the alternative hypothesis that the average CGPA of undergraduate students is less than 8. What type of test should be used, and what is the critical value at $\alpha = 5\%$?

a. One-tailed test and $+1.96$
b. One-tailed test and -1.96

c. One-tailed test and $+1.645$
d. One-tailed test and -1.645

See Section 2.8.

QB 11.3 The margin of error for a confidence interval is $\pm Z_{\alpha/2}\,\sigma/\sqrt{n}$. Which statements are correct? *(select all that apply)*

a. Increasing the sample size will decrease the margin of error
b. The margin of error decreases as α rises
c. The larger the margin of error, the less confident one is about the estimate
d. The margin of error increases with the population standard deviation σ

See Section 2.6.

QB 11.4 A 95% confidence interval for mean student height is $(165.41,\ 170.95)$. You now compute a 99% interval from the same data. Which statements are likely correct? *(select all that apply)*

a. The 90%, 95% and 99% intervals are the same
b. The 99% interval is $(164.54,\ 171.83)$
c. The 99% interval is $(165.86,\ 170.51)$
d. The answer cannot be determined from the information given

See Section 2.6. Ask yourself which way an interval must move when you demand more confidence.

QB 11.5 A company tests the null hypothesis that the average time to accept a job offer is at most 7 days. The test concludes that the average exceeds 7 days. It is later learned that the true mean is exactly 7 days. What error occurred?

a. Type II error
b. Type I error
c. A correct decision was made
d. Cannot be determined from the information provided

Short reasoning

QB 11.6 "A high p-value is evidence that the null hypothesis is true." Do you agree? Why or why not?

See Section 2.5.

QB 11.7 Two economists, Amit and Anand, collect data on Indian agricultural wages to test the effect of MNREGA. Amit concludes that wages after the policy are higher than before, when in fact wages have not changed. Anand concludes that wages before and after are the same — also incorrectly.

It is claimed that Amit has committed a Type I error and Anand a Type II error. Do you agree? Why or why not?

Numerical

QB 11.8 A researcher wants to estimate the average weekly study hours of university students. The population standard deviation is 6 hours. What sample size is needed to estimate the mean with a margin of error of 1 hour at 90% confidence?

a. $n = 98$
b. $n = 64$
c. $n = 121$
d. $n = 82$

QB 11.9 The numbers of people taking the morning university shuttle on 16 randomly chosen days are:

```
27, 46, 35, 33, 29, 45, 28, 24, 30, 40, 38, 35, 41, 27, 38, 29
```

The administration claims the daily average is 30. Taking $\sigma = 7$:

a. Construct a 99% confidence interval for the mean number of riders.
b. Use that interval to test the administration's claim.
c. Would your conclusion change at the 95% level? Explain *why* before you compute it.

Part (c) is the one that separates understanding from arithmetic.

2.12 Summary

REMEMBER THIS

1. A gap between a sample and a claim proves nothing on its own — sampling noise guarantees some gap. The test asks whether the gap is *bigger than noise*.

2. $Z_{calc} = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ when σ is known; $t_{calc} = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ with $df = n-1$ when it is not.
3. Reject when the statistic beats the critical value, or equivalently when the p-value falls below α. The two rules never disagree.
4. A confidence interval is the set of null values a test would *not* reject. Interval and test are the same statement.
5. A p-value is $P(\text{data} \mid H_0)$, never $P(H_0 \mid \text{data})$.
6. The equals sign always lives in H_0. "Can you reject the claim?" puts the claim in H_0; "is there evidence supporting the claim?" puts it in H_A.
7. `qnorm()` and `qt()` replace every printed table you own.

Next we extend all of this from one sample to two, and ask whether *two groups* differ — the question behind almost every policy evaluation you will ever read.